

Haohan Wang

Assistant Professor

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Research Interests

Computational Biology	Building Virtual Bioinformaticians to Accelerate Scientific Discovery Process Computational Methods to Understand Regulatory Structure from Spatial Transcriptomics
Agentic A.I.	Multi-agent Systems to Automate Complicated Real-world Tasks, especially in AI4Science Scope Statistical Understanding of Multi-agent System
Trustworthy A.I.	Defending LLM-driven AI against Potentially Dangerous Use Cases

I Appointments and Positions

2022 -	Assistant Professor, School of Information Sciences University of Illinois Urbana-Champaign
2023 -	Affiliated Faculty, Department of Computer Science University of Illinois Urbana-Champaign
2023 -	Affiliated Faculty, National Center for Supercomputing Applications University of Illinois Urbana-Champaign
2023 -	Affiliated Faculty, Gene Networks in Neural & Developmental Plasticity Carl R. Woese Institute for Genomic Biology, UIUC
2022 - 2022	Postdoctoral Researcher, Machine Learning Department Carnegie Mellon University
2021 -	Organizer Trustworthy Machine Learning Initiative

II Education

2015 - 2021	Ph.D, Language Technologies Institute, Carnegie Mellon University Advisor: Eric P. Xing
2012 - 2015	M.S. Language Technologies Institute, Carnegie Mellon University Advisor: Madhavi Ganapathiraju
2008 - 2012	B.S. Computer Science, Beijing Univ. of Posts & Telecoms.

III Awards & Research Support

- Haohan Wang, PI., Toward the Next Generation Recommendation with Large Language Models, from US Bank. \$25,000, unrestricted gift, 2026

- Haohan Wang, PI., Task-Grounded Security for LLMs Toward Secure Instruction Following and Mitigating Prompt Injections, from National Artificial Intelligence Research Resource (NAIRR) Pilot, \$10,000 (as OpenAI API credits) 2025.
- Haohan Wang, PI., Adaptive Representation Guard: Detecting and Neutralizing Prompt Injection via LLM Activation Monitoring, Amazon-Illinois Center on AI for Interactive Conversational Experiences (AICE) 2025-2026, \$72,982
- Haohan Wang, PI., Toward Virtual Bioinformaticians: Empowering Scientific Discovery in Genomics with LLM-Based Agents, from National Artificial Intelligence Research Resource (NAIRR) Pilot, \$90,000 (as computing credits: \$50,000 OpenAI credits, \$30,000 Amazon AWS credits, and \$10,000 from HuggingFace)
- Haohan Wang, PI., Toward Deep Learning Techniques for Cell-Type and Spatial Resolution Estimation of Regulatory Networks, from National Institutes of Health, 2024-2026, \$290,326
- Haohan Wang, PI., Predicting Vasovagal Syncope in Postural Orthostatic Tachycardia Syndrome Patients using Machine Learning Methods and Interpretation of Syncopal Episodes. Standing Up to POTS, 2024-2027, \$38,800
- Haohan Wang, PI., Microsoft Accelerating Foundation Models Research, 2023, \$25,000 (as computing credits)

IV Recognitions

- Travel Support Award from Pacific Symposium on Biocomputing.
- Outstanding Paper Award at FL@FM workshop at NeurIPS 2024
- Best Paper - Honorable Mention in CIKM 2023
- Nomination of KDD Dissertation Award by School of Computer Science, CMU, 2022
- Rising Stars as Young Investigators in AI+X by Baidu Research, 2022
- Youth Outstanding Paper Award by The World Artificial Intelligence Conference, 2021
- Youth Outstanding Paper Award by The World Artificial Intelligence Conference, 2020
- Next Generation in Biomedicine by the Broad Institute of MIT and Harvard, Sept. 2019
- Fellowship Award (Finalist) by Baidu Research, 2019
- Most Influential Data Science Research Papers in 2018 by media from Open Data Science Conference (ODSC), Dec. 2018
- Student Paper Award by IEEE BIBM. Nov. 2017

V Publications

Journal Publications

- [1]. Rao H, Liu W, **Wang H**, Huang I-C, He Z, Huang X. A Scoping Review of Synthetic Data Generation for Biomedical Research and Applications. *Journal of Healthcare Informatics Research* 2026.
- [2]. Chen K, Wang P, Yu Y, Zhan X, **Wang H**. Large Language Model-based Data Science Agent: A Survey. *Transactions of Machine Learning Research (TMLR)* 2026.
- [3]. Deng C, Duan Y, Jin X, Chang H, Tian Y, Liu H, Wang Y, Gao K, Zou HP, Jin Y, Xiao Y, Wu S, Xie Z, Lyu W, He S, Cheng L, **Wang H**, Zhuang J. Deconstructing the ethics of large language models from long-standing issues to new-emerging dilemmas: a survey. *AI and Ethics*. 2025 Aug 13;5:4745–4771.

- [4]. Li Y, Gao Y, **Wang H**. Understanding Adversarial Transferability in Federated Learning. Transactions of Machine Learning Research (TMLR) 2024.
- [5]. Anzaku ET, **Wang H**, Van Messem A, De Neve W, Re-assessing Accuracy Degradation: A Framework for Understanding DNN Behavior on Similar-but-non-identical Test Datasets. ACML journal track 2024.
- [6]. Leng J, Li Y, **Wang H**. Choosing Wisely and Learning Deeply: Selective Cross-Modality Distillation via CLIP for Domain Generalization. Transactions of Machine Learning Research (TMLR) 2024.
- [7]. Leng J, Liu J, Cheng G, **Wang H**, Quarrier S, Luo J, Jain R. Development of UroSAM: a machine learning model to automatically identify kidney stone composition from endoscopic video. Journal of Endourology. 2024 May 31.
- [8]. Miranda O, Fan P, Qi X, **Wang H**, Brannock MD, Kosten TR, Ryan ND, Kirisci L, Wang L. DeepBiomarker2: Prediction of alcohol and substance use disorder risk in post-traumatic stress disorder patients using electronic medical records and multiple social determinants of health. Journal of Personalized Medicine. 2024 Jan 14;14(1):94.
- [9]. Zhang J, Song B, **Wang H**, Han B, Liu T, Liu L, Sugiyama M. BadLabel: A Robust Perspective on Evaluating and Enhancing Label-Noise Learning. IEEE Transactions on Pattern Analysis and Machine Intelligence. 2024 Jan 18.
- [10]. Miranda O, Fan P, Qi X, **Wang H**, Brannock MD, Kosten T, Ryan ND, Kirisci L, Wang L. Prediction of Adverse Events Risk in Patients with Comorbid Post-Traumatic Stress Disorder and Alcohol Use Disorder Using Electronic Medical Records by Deep Learning Models. Drug and Alcohol Dependence. 2024 Jan 9;111066.
- [11]. Lee, K.Y., **Wang, H.**, Yook, Y., Rhodes, J.S., Christian-Hinman, C.A. and Tsai, N.P., 2023. Tumor suppressor p53 modulates activity-dependent synapse strengthening, autism-like behavior and hippocampus-dependent learning. Molecular Psychiatry, pp.1-13.
- [12]. **Wang H**, Lopez O, Xing EP, Wu W. Kernel Mixed Model for Transcriptome Association Study. Journal of Computational Biology. 2022 Dec 1;29(12):1353-6.
- [13]. **Wang H**, Lopez OL, Wu W, Xing EP. Gene set prioritization guided by regulatory networks with p-values through kernel mixed model. In Journal of Computational Biology 2022
- [14]. Miranda O, Fan P, Qi X, Yu Z, Ying J, **Wang H**, Brent DA, Silverstein JC, Chen Y, Wang L. DeepBiomarker: identifying important lab tests from electronic medical records for the prediction of suicide-related events among PTSD patients. Journal of personalized medicine. 2022 Mar 24;12(4):524.
- [15]. **Wang H**, Pei F, Vanyukov MM, Bahar I, Wu W, Xing EP. Coupled mixed model for joint genetic analysis of complex disorders with two independently collected data sets. BMC bioinformatics. 2021 Dec;22(1):1-4.
- [16]. **Wang H**, Aragam B, Xing EP. Trade-offs of Linear Mixed Models in Genome-Wide Association Studies. Journal of Computational Biology. 2022 Mar 1;29(3):233-42.
- [17]. Du X, **Wang H**, Zhu Z, Zeng X, Chang YW, Zhang J, Xing E, Xu M. Active learning to classify macromolecular structures in situ for less supervision in cryo-electron tomography. Bioinformatics. 2021 Aug 25;37(16):2340-6.
- [18]. Ge S, **Wang H**, Alavi A, Xing E, Bar-Joseph Z. Supervised adversarial alignment of single-cell RNA-seq data. Journal of Computational Biology. 2021 May 1;28(5):501-13.
- [19]. Zheng Y, **Wang H**, Zhang Y, Gao X, Xing EP, Xu M. Poly (A)-DG: A deep-learning-based domain generalization method to identify cross-species Poly (A) signal without prior knowl-

- edge from target species. PLoS computational biology. 2020 Nov 5;16(11):e1008297.
- [20]. **Wang H**, Vanyukov MM, Xing EP, Wu W. Discovering weaker genetic associations guided by known associations. BMC Medical Genomics. 2020 Feb;13:1-0.
 - [21]. **Wang H**, Yue T, Yang J, Wu W, Xing EP. Deep mixed model for marginal epistasis detection and population stratification correction in genome-wide association studies. BMC bioinformatics. 2019 Dec;20:1-1.
 - [22]. Xu M, Wu J, **Wang H**, Cao M. Anomaly detection in road networks using sliding-window tensor factorization. IEEE Transactions on Intelligent Transportation Systems. 2019 Sep 25;20(12):4704-13.
 - [23]. Xiao Y, Chen D, Wei S, Li Q, **Wang H**, Xu M. Rumor propagation dynamic model based on evolutionary game and anti-rumor. Nonlinear Dynamics. 2019 Jan;95:523-39.
 - [24]. Xu M, Wu J, Liu M, Xiao Y, **Wang H**, Hu D. Discovery of critical nodes in road networks through mining from vehicle trajectories. IEEE Transactions on Intelligent Transportation Systems. 2018 May 11;20(2):583-93.
 - [25]. **Wang H**, Lengerich BJ, Aragam B, Xing EP. Precision Lasso: accounting for correlations and linear dependencies in high-dimensional genomic data. Bioinformatics. 2019 Apr 1;35(7):1181-7.
 - [26]. **Wang H**, Aragam B, Xing EP. Variable selection in heterogeneous datasets: A truncated-rank sparse linear mixed model with applications to genome-wide association studies. Methods. 2018;145:2-9.
 - [27]. Lee S, **Wang H**, Xing EP. Backward genotype-transcript-phenotype association mapping. Methods. 2017 Oct 1;129:18-23.

Conference Publications

- [28]. Yu Y, Zhang P, Yu Y, Wei K, Luo H, **Wang H**. SIPDO: Closed-Loop Prompt Optimization via Synthetic Data Feedback. In Proceedings of the International Conference on Learning Representations (ICLR), 2026.
- [29]. Kuang P, Wang Y, Han X, Liu Y, Xu K, **Wang H**. Optimal Aggregation of LLM and PRM Signals for Efficient Test-Time Scaling. In Proceedings of the International Conference on Learning Representations (ICLR), 2026.
- [30]. Chen K, Yu X, Zhou Y, **Wang H**. Prompt Stability Matters: Evaluating and Optimizing Auto-Generated Prompt in General-Purpose Systems. In Proceedings of the Conference on Parsimony and Learning (CPAL), 2026.
- [31]. Yu Y, Jin H, Yu Y, Zhuang J, **Wang H**. Now You Hear Me: Audio Narrative Attacks Against Large Audio–Language Models. In Proceedings of the European Chapter of the Association for Computational Linguistics (EACL), 2026.
- [32]. Liu H, Li Y, **Wang H**. GenoMAS: A Multi-Agent Framework for Scientific Discovery via Code-Driven Gene Expression Analysis. In Proceedings of the European Chapter of the Association for Computational Linguistics (EACL), 2026.
- [33]. Guan J, Jin H, **Wang H**. Crafting Adversarial Inputs for Large Vision-Language Models Using Black-Box Optimization. In Proceedings of the European Chapter of the Association for Computational Linguistics (EACL), 2026.
- [34]. Li Z, Yuan X, Shen B, Le K, **Wang H**, Zhou X, Gao S, Dong Y. Query-efficient domain knowledge stealing against large language models. In Proceedings of the AAAI Conference on Artificial Intelligence (AAAI) 2026.

- [35]. Chen K, **Wang H**. Discovery of disease relationships via transcriptomic signature analysis powered by agentic AI. In Proceedings of the Pacific Symposium on Biocomputing (PSB) 2026.
- [36]. Jin H, Zhang P, Luo M, **Wang H**. Reasoning can hurt the inductive abilities of large language models. In Advances in Neural Information Processing Systems (NeurIPS) 2025.
- [37]. Zhuang J, Jin H, Zhang Y, Kang Z, Zhang W, Dagher G G, **Wang H**. Exploring the Vulnerability of the Content Moderation Guardrail in Large Language Models via Intent Manipulation. EMNLP Findings 2025.
- [38]. Xue Z, **Wang H**, Qin Y, Pedarsani R. Conflict-Aware Adversarial Training. BMVC 2025
- [39]. Liu H, Chen S, Zhang Y, **Wang H**. GenoTEX: An LLM Agent Benchmark for Automated Gene Expression Data Analysis. In proceedings of Machine Learning for Computational Biology 2025
- [40]. Leng J, Huang C, Huang L, Lin B Y, Cohen W W, **Wang H**, Huang J. CrossWordBench: Evaluating the Reasoning Capabilities of LLMs and LVLMs with Controllable Puzzle Generation. COLM 2025
- [41]. Liu H, Li Y, Xing T, Wang P, Dalal V, Li L, He J, **Wang H**. Dataset Distillation via the Wasserstein Metric. ICCV 2025
- [42]. Huang Z, Ji Y, **Wang H**, Lee Y. Customizing Domain Adapters for Domain Generalization. ICCV 2025
- [43]. Zheng R, Dasu V A, Wang Y O, **Wang H**, De la Torre F. Improving Noise Efficiency in Privacy-preserving Dataset Distillation. ICCV 2025
- [44]. Xu Y, Nallamothu S, Culbertson C, **Wang H**. Privacy-preserving analysis of wearable device data for monitoring health risks. In Proceedings of the 2025 IEEE 13th International Conference on Healthcare Informatics (ICHI) 2025 Jun (pp. 708-715). IEEE.
- [45]. Zhang P, Jin H, Hu L, Li X, Kang L, Luo M, Song Y, **Wang H**. Revolve: Optimizing AI Systems by Tracking Response Evolution in Textual Optimization. ICML 2025
- [46]. Jeoung S, Ge Y, **Wang H**, Diesner J. Examining Alignment of Large Language Models through Representative Heuristics: The Case of Political Stereotypes. ICLR 2025
- [47]. Liu H, Singh A, Li Y, **Wang H**. Approximate Nullspace Augmented Finetuning for Robust Vision Transformers. Conference on Parsimony and Learning (CPAL) 2025
- [48]. Xue E, Li Y, Liu H, Wang P, Shen Y, **Wang H**. Towards adversarially robust dataset distillation by curvature regularization. AAAI 2025.
- [49]. Cai M, Huang Z, Li Y, Ojha U, **Wang H**, Lee YJ. Leveraging large language models for scalable vector graphics-driven image understanding. WACV 2025.
- [50]. Zhou A, Li B, **Wang H**. Robust prompt optimization for defending language models against jailbreaking attacks. in proceedings of NeurIPS 2024
- [51]. Jin H, Zhou A, Menke JD, **Wang H**. Jailbreaking Large Language Models Against Moderation Guardrails via Cipher Characters. in proceedings of NeurIPS 2024
- [52]. Shah S.B, Shiwakoti S, Chaudhary M, **Wang H**. MemeCLIP: Leveraging CLIP Representations for Multimodal Meme Classification. EMNLP 2024
- [53]. Fei Y, Li Y, Zhou Z, **Wang H**. DiffSim: Aligning Diffusion Model and Molecular Dynamics Simulation for Blind Docking. BIBM 2024
- [54]. Tam TY, Liang L, Chen K, **Wang H**, Wu W. A Quantitative Approach for Evaluating Disease Focus and Interpretability of Deep Learning Models for Alzheimer’s Disease Classification. BIBM 2024

- [55]. Singh A, **Wang H**. Simple Unsupervised Knowledge Distillation With Space Similarity. ECCV 2024
- [56]. Chen R, Jin H, Liu Y, Chen J, **Wang H**, Sun L. EditShield: Protecting Unauthorized Image Editing by Instruction-guided Diffusion Models. ECCV 2024.
- [57]. Jin H, Chen R, Chen J, Zheng H, Zhang Y, **Wang H**, CatchBackdoor: Backdoor Testing by Critical Trojan Neural Path Identification via Differential Fuzzing. ECCV 2024.
- [58]. Chen H, Zhuang J, Yao Y, Jin W, **Wang H**, Xie Y, Chi CH, Choo KK. Trustworthy and Responsible AI for Information and Knowledge Management System. In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management 2024 Oct 21 (pp. 5574-5576).
- [59]. Zhou A, Yan K, Shlapentokh-Rothman M, **Wang H**, Wang YX. Language agent tree search unifies reasoning acting and planning in language models. ICML 2024.
- [60]. Zhang, P., Liu, H., Li, C., Xie, X., Kim, S. and **Wang, H.**, Foundation model-oriented robustness: Robust image model evaluation with pretrained models. ICLR 2024
- [61]. Zhou A, Wang J, Wang YX, **Wang H**. Distilling Out-of-Distribution Robustness from Vision-Language Foundation Models. NeurIPS 2023.
- [62]. Bao, W., Wei, T., **Wang H**. and He, J., Adaptive Test-Time Personalization for Federated Learning. NeurIPS 2023.
- [63]. Wang J, Li H, **Wang H**, Pan SJ, Xie X. Trustworthy machine learning: Robustness, generalization, and interpretability. In Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining 2023 Aug 6 (pp. 5827-5828).
- [64]. Huang, Z., Zhou, A., Lin, Z., Cai, Mu., Lee, YJ.*, **Wang, H.*** A Sentence Speaks a Thousand Images: Domain Generalization through Distilling CLIP with Language Guidance, in proceedings of 2023 International Conference on Computer Vision. ICCV 2023 * equal advising role.
- [65]. Bao, W., **Wang, H.**, Wu, J., and He, J. Optimizing the Collaboration Structure in Cross-Silo Federated Learning , in proceedings of 2023 International Conference on Machine Learning. ICML 2023
- [66]. Zhang P, Guo J, Li C, Xie Y, Kim JB, Zhang Y, Xie X, **Wang H**, Kim S. Efficiently leveraging multi-level user intent for session-based recommendation via atten-mixer network. In Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining 2023 Feb 27 (pp. 168-176). **Best Paper - Honorable Mention**
- [67]. Xiang K, Zhang X, She J, Liu J, **Wang H**, Deng S, Jiang S. Toward Robust Diagnosis: A Contour Attention Preserving Adversarial Defense for COVID-19 Detection. AAAI 2023.
- [68]. **Wang H**, Huang Z, Wu X, Xing E. Toward learning robust and invariant representations with alignment regularization and data augmentation. In Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining 2022 Aug 14 (pp. 1846-1856).
- [69]. **Wang H***, Huang Z*, Huang D, Lee YJ, Xing EP. The two dimensions of worst-case training and their integrated effect for out-of-domain generalization. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition 2022 (pp. 9631-9641). * equal contribution.
- [70]. **Wang H**, Huang Z, Zhang H, Lee YJ, Xing EP. Toward learning human-aligned cross-domain robust models by countering misaligned features. In Uncertainty in Artificial Intelligence 2022 Aug 17 (pp. 2075-2084). PMLR.
- [71]. Wang X, **Wang H**, Yang D. Measure and improve robustness in nlp models: A survey.

- [72]. Ge S, Mishra S, Li CL, **Wang H**, Jacobs D. Robust contrastive learning using negative samples with diminished semantics. *Advances in Neural Information Processing Systems*. 2021 Dec 6;34:27356-68.
- [73]. **Wang H***, Huang Z*, Xing EP, Huang D. Self-challenging improves cross-domain generalization. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part II* 16 2020 (pp. 124-140). Springer International Publishing. * equal contribution. **Oral**
- [74]. **Wang H**, Wu X, Huang Z, Xing EP. High-frequency component helps explain the generalization of convolutional neural networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition 2020* (pp. 8684-8694). **Oral**
- [75]. **Wang H**, Lu C, Wu W, Xing EP. Graph-structured sparse mixed models for genetic association with confounding factors correction. In *2019 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* 2019 Nov 18 (pp. 298-302). IEEE.
- [76]. **Wang H**, Wei Y, Cao M, Xu M, Wu W, Xing EP. Deep inductive matrix completion for biomedical interaction prediction. In *2019 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* 2019 Nov 18 (pp. 520-527). IEEE.
- [77]. Wu X, Mao Y, **Wang H**, Zeng X, Gao X, Xing EP, Xu M. Regularized adversarial training (RAT) for robust cellular electron cryo tomograms classification. In *2019 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* 2019 Nov 18 (pp. 1-6). IEEE.
- [78]. **Wang H**, Sun D, Xing EP. What if we simply swap the two text fragments? A straightforward yet effective way to test the robustness of methods to confounding signals in nature language inference tasks. In *Proceedings of the AAAI Conference on Artificial Intelligence* 2019 Jul 17 (Vol. 33, No. 01, pp. 7136-7143).
- [79]. **Wang H**, Xing EP, He Z, Lipton ZC. Learning robust representations by projecting superficial statistics out. In *7th International Conference on Learning Representations, ICLR 2019* 2019. **Oral**
- [80]. **Wang H**, Ge S, Lipton Z, Xing EP. Learning robust global representations by penalizing local predictive power. *Advances in Neural Information Processing Systems*. 2019;32.
- [81]. **Wang H**, Wu Z, Xing EP. Removing confounding factors associated weights in deep neural networks improves the prediction accuracy for healthcare applications. In *BIOCOMPUTING 2019: Proceedings of the Pacific Symposium 2018* (pp. 54-65). **Oral**
- [82]. **Wang H**, Liu X, Tao Y, Ye W, Jin Q, Cohen WW, Xing EP. Automatic human-like mining and constructing reliable genetic association database with deep reinforcement learning. In *BIOCOMPUTING 2019: Proceedings of the Pacific Symposium 2018* (pp. 112-123).
- [83]. **Wang H**, Liu X, Xiao Y, Xu M, Xing EP. Multiplex confounding factor correction for genomic association mapping with squared sparse linear mixed model. In *2017 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* 2017 Nov 13 (pp. 194-201). IEEE. **Student Paper Award**
- [84]. **Wang H**, Meghawat A, Morency LP, Xing EP. Select-additive learning: Improving generalization in multimodal sentiment analysis. In *2017 IEEE International Conference on Multimedia and Expo (ICME)* 2017 Jul 10 (pp. 949-954). IEEE.
- [85]. **Wang H**, Yang J. Multiple confounders correction with regularized linear mixed effect models, with application in biological processes. In *2016 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* 2016 Dec 15 (pp. 1561-1568). IEEE.

- [86]. Gupta A, **Wang H**, Ganapathiraju M. Learning structure in gene expression data using deep architectures, with an application to gene clustering. In 2015 IEEE international conference on bioinformatics and biomedicine (BIBM) 2015 Nov 9 (pp. 1328-1335). IEEE.
- [87]. **Wang H**, Raza AA, Lin Y, Rosenfeld R. Behavior analysis of low-literate users of a viral speech-based telephone service. In Proceedings of the 4th Annual Symposium on Computing for Development 2013 Dec 6 (pp. 1-9).

Selective Conference Workshop Publications

- [88]. Xu Y, Nallamothu S, Liu H, **Wang H**. GenePrep: Unified Genomic Data Formatter for Statistical Analysis. Machine Learning for Computational Biology 2025 (poster)
- [89]. Wang L, Yu H, Wang J, Teney D, **Wang H.**, Zhu Y, Chen Y, Yang Q, Xie X, Ji X. ZOOPFL: Exploring Black-box Foundation Models for Personalized Federated Learning. FL@FM workshop at NeurIPS'24 **Outstanding Paper Award**.
- [90]. Zhang Y*, **Wang H**. Photong: Generating 16-Bar Melodies from Images. In The AAAI-23 Workshop on Creative AI Across Modalities 2023. * **K-12 author**
- [91]. **Wang H**, Chen C, Hu L, Zhang Y, Lyu S, Wu J, Li X, Sun L, Xing EP. Robustar: Interactive Toolbox Supporting Precise Data Annotation for Robust Vision Learning. ICML DataPerf Workshop 2022
- [92]. He H, Zha S, **Wang H**. Unlearn dataset bias in natural language inference by fitting the residual. EMNLP Workshop 2019
- [93]. Moon S, Kim S, **Wang H**. Multimodal transfer deep learning for au-dio visual recognition. NeurIPS workshop 2014
- [94]. **Wang H**, Li Y, Hu X, Yang Y, Meng Z, Chang KM. Using EEG to Improve Massive Open Online Courses Feedback Interaction. In AIED Workshops 2013 Jun.

Selective Technical Reports

- [95]. Yadav A, Jin H, Luo M, Zhuang J, **Wang H**. InfoFlood: Jailbreaking Large Language Models with Information Overload. arXiv preprint arXiv:2506.12274. 2025. **Reported by media such as 404 Media, POLITICO, IT Brew, Illinois News Bureau**
- [96]. Jin H, Hu L, Li X, Zhang P, Chen C, Zhuang J, **Wang H**. JailbreakZoo: Survey, Landscapes, and Horizons in Jailbreaking Large Language and Vision-Language Models. arXiv preprint arXiv:2407.01599. 2024 Jun 26.
- [97]. Liu H, Chaudhary M, **Wang H**. Towards trustworthy and aligned machine learning: A data-centric survey with causality perspectives. arXiv preprint arXiv:2307.16851. 2023 Jul 31.
- [98]. Yue T, **Wang H**. Deep learning for genomics: A concise overview. arXiv preprint arXiv:1802.00810. 2018 Feb 2.
- [99]. **Wang H**, Raj B. On the origin of deep learning. arXiv preprint arXiv:1702.07800. 2017 Feb 24.

VI Invited Presentations & Talks

- **Practical Concerns of Prompt Injection in Deployed Language Models**
Capitol One
Virtual, Dec. 2025
- **Agentic AI for Genomic Science: Understanding the Mechanism of Thousands of Diseases at One Time**

- Data Science Lecture at William & Mary
Virtual, Nov. 2025
- **Large Language Model-based Data Science Agent**
ResearchTrend.AI
Virtual, Nov. 2025
 - **Detecting and Neutralizing Prompt Injection via LLM Activation Monitoring**
AICE Center
UIUC, Oct. 2025
 - **Toward Agentic AI Scientist for Biomedical Discovery**
Tsinghua University
Virtual, Aug. 2025
 - **Large Language Models Struggles in Induction, and Reasoning Makes It Worse**
ProSem Lecture Series, iSchool
UIUC, April 2025
 - **Spatially Varying Cell-specific Gene Regulation Network Inference**
NIH Common Fund Data Ecosystem (CFDE) meeting
Bethesda, Mar 2025
 - **Toward Agentic AI Scientist for Biomedical Discovery**
Generative AI and the Future of Research” CIRSS speaker series talk
UIUC, Mar 2025
 - **GenoAgent: LLM-Based Exploration of Gene Expression Data in Alignment with Bioinformaticians**
NAIRR Pilot
Washington, D.C. Feb. 2025
 - **Toward Agentic AI Scientist for Biomedical Discovery**
Reading Group from Australian National University
virtual, Jan 2025
 - **AI Powered Transformations in Finance, Toward Underwriting Automation**
Southwestern Ohio Winter Meeting,
Cincinnati, Dec. 2024
 - **Guardian of Trust in Language Models: Automatic Jailbreak and Systematic Defense**
Computer Science Seminar at University of Illinois, Chicago
virtual, July 2024
 - **Towards LLM-Based Exploration of Gene Expression Data in Alignment with Bioinformaticians**
NIH/NLM
virtual, July. 2024
 - **Understanding Variations in Regulatory Networks Across Cell Types through Transformer Model with Knowledge on Regulatory Interactions**
Genetics and Epigenetics Cross-Cutting Research Team Meeting, National Institute on Drug Abuse (NIDA), NIH Campus
Bethesda, Maryland, Jun. 2024
 - **Technical Advances of Jailbreaks**
Analytics Lunch & Learn at Protiviti (a consulting firm)
virtual, May 2024,

- **Guardian of Trust in Language Models: Automatic Jailbreak and Systematic Defense**
Trustworthiness of Large Language Models, CS@Virgia Tech
Virtual, April. 2024
- **A Team of AI-made Scientists from LLMs for Scientific Discovery from Transcriptomics**
Recent developments in large language models for biology, EMBL-EBI Workshop
Boston, April. 2024
- **Guardian of Trust in Language Models: Automatic Jailbreak and Systematic Defense**
AI Talks
Virtual, April. 2024
- **Guardian of Trust in Language Models: Automatic Jailbreak and Systematic Defense**
Large Language Model Reading Group, UIUC
Illinois, March. 2024
- **Advancing Precision Medicine: Tailored Genomic Insights and AI-Driven Automation in Complex Disease Research**
Computer Science Department, UIUC
Illinois, March. 2024
- **Advancing Precision Medicine: Tailored Genomic Insights and AI-Driven Automation in Complex Disease Research**
Biostatistics & Medical Informatics Seminar, University of Wisconsin-Madison
Madison, Feb. 2024
- **Deep Learning Methods to Navigate Heterogeneous Data Landscapes for Genetic Insights**
Spatial Genomics Seminar, Carl R. Woese Institute for Genomic Biology, UIUC
Champaign, Feb. 2024
- **Guardian of Trust in Language Models: Automatic Jailbreak and Systematic Defense**
VALSE: Vision and Learning Seminar
virtual, Jan. 2024
- **Understanding Structural Patterns for Early-Diagnosis of Alzheimer's Disease**
Stanford AI+Health Seminar
virtual, Dec. 2023
- **Role Playing to Improve the Trustworthiness of Large Language Models**
Champaign-Urbana Data Science User Group
Champaign IL. Oct. 2023
- **Tutorial in Trustworthy Machine Learning**
2023 ACM SIGKDD International Conference on Knowledge Discovery and Data Mining
Long Beach CA. Aug. 2023
- **Robust Machine Learning Techniques Helps Understand Alzheimer's Disease Subtypes**
HealthX Club of Shanghai Jiao Tong University Seminar
virtual, July. 2023
- **Tutorial in Trustworthy Machine Learning for Biomeical Challenges**
2023 IEEE International Conference on Healthcare Informatics
Houston, Texas. June. 2023
- **Robust Machine Learning with Big Models**
Halil Lab, iSchool, UIUC
Champaign IL. Dec. 2022
- **Toward a Principled Understanding of Trustworthy Methods Machine Learning**

- “Xia Peisu Young Scholars Forum” at Chinese Academy of Sciences Institute of Computing Technology,
virtual, Dec. 2022
- **Causality and Multiple Aspects of Trustworthy Machine Learning**
Keynote, TrustLog WorkShop at CIKM
Atlanta, Georgia. Oct. 2022
 - **Toward the Alignment Between Machine Learning Models’ and Users’ Perceptions of Data**
Champaign-Urbana Data Science User Group
Champaign IL. Sept. 2022
 - **Toward a Principled Understanding of Robust ML and Its Connection to Multiple Aspects**
Doctoral ProSeminar at iSchool, UIUC
Champaign IL. Sept. 2022
 - **Toward a Principled Understanding of Robust Machine Learning Methods and Its Connection to Multiple Aspects**
Seminar of Machine Learning and NLP (MLNLP seminar series)
virtual Aug. 2022
 - **Trustworthy AI-diagnosis of Alzheimer’s Disease from MRI**
CNS Lab, Stanford University
virtual July. 2022
 - **A Principled Understanding of Robust Machine Learning Methods**
RIKEN Center for Advanced Intelligence Project, Japan
virtual July. 2022
 - **Robust Computer Vision Techniques Help Identify Subtypes of Alzheimer’s Disease**
Department of Biomedical Informatics Colloquium Series, University of Pittsburgh
Pittsburgh PA. Dec. 2021
 - **Robust Machine Learning with Emphasis on Countering Spurious Features**
Data Science Initiative, Brown University
virtual. March. 2021
 - **High Frequency Component Helps Explain the Generalization of CNN**
Aggregate Intellect
virtual. Jan., 2021.
 - **Towards Trustworthy Machine Learning Inspired by High-frequency Image Data**
R-Pad meeting, Robotics Institute, Carnegie Mellon University.
virtual. Nov., 2020.
 - **Towards Trustworthy Machine Learning for Scientific Discovery**
Doctoral Symposium at ACM Conference on Health, Inference, and Learning.
virtual. July, 2020.
 - **Learning Deconfounded Representations through Neural Networks, with Applications in Genetic Data**
Center of Excellence for Computational Drug Abuse Research,
Pittsburgh, USA. Feb., 2020.
 - **Dealing with Confounding Factors with Deep Neural Networks**
Next Generation in Biomedicine Symposium, the Broad Institute.
Boston, USA. Sept. 2019.
 - **Deep Learning over Heterogeneous Data**

Department of Biomedical Informatics Colloquium Series, University of Pittsburgh.
Pittsburgh, USA. Sept. 2019.

- **Learning Robust Representations by Projecting Superficial Statistics Out**

Machine Learning Lunch Seminar, Carnegie Mellon University.
Pittsburgh, USA. Feb. 2019.

VII Teaching

- IS507 Data, Statistical Models, and Information
Fall 2023, Fall 2024, Fall 2025
- IS557 Applied Machine Learning Team Projects
Spring 2023
- IS597 Trustworthy Machine Learning
Spring 2024, Spring 2025

VIII Community Service

- **Organizer:**
 - The Midwest ML Symposium 2026 (as general chair)
 - DCAI Workshop at Web Conference (WWW) 2024
 - Trustworthy Machine Learning workshop at CIKM 2024
- **Journal Editor:** Nature Scientific Report
- **Vertical Chair:** AI Standard 2025
- **Area Chair:** ICML 2024, 2025, 2026, ICLR AGI Workshop 2024, ACL 2025, AAAI (as SPC) 2024-2025
- **Journal Reviewer:** PNAS, Nature Scientific Data, Nature Machine Intelligence, IEEE Transactions on Cognitive and Developmental Systems, PLOS Computational Biology, Bioinformatics, BMC Bioinformatics, etc.
- **Conference Reviewer (Program Committee):** ICML, NeurIPS, ICLR, CVPR, ICCV, UAI, AISTATS, MLCB, BIBM, ACL, EACL, NAACL, etc.
- **Panelist**
 - NSF-NIH Joint Workshop in Emerging AI in Biology
 - NIH Review Panel April. 2025
 - NIH Review Panel July. 2025
 - NIH Review Panel Oct. 2025
 - NIH Review Panel Jan. 2026
 - NIH Review Panel Jan. 2026
 - NSF Review Panel Jan. 2026
- **Mentor:** NIH CFDE program summer 2025
- **International Mentor:** Deep Learning Indaba Mentorship for African Scholars; UIUC Bolashak Data Science Program for Bolashak Scholars.